1. Clinical Question & Core Hypothesis

Clinical Question

Cisplatin both damages DNA (increasing mutational/neoantigen burden) and selects for TP53-deficient clones; pre-existing organized immune structures (TLS) should promote response to PD-1 blockade, yet clinical benefit of pembrolizumab+cisplatin is variable. Why do some TLS-rich muscle-invasive urothelial tumors fail to respond when exposed to cisplatin plus PD-1 blockade? 2) Tier 1 paths used: (a) TP53 mutation status as a core modifier of DNA-damage responses and clonal evolution; (b) cisplatin-induced mutational processes that change tumor antigenicity. I built on these because they provide a direct mechanistic link from chemotherapy to neoantigen landscape and immune recognition. 3) Tier 2 supporting evidence: increased overall tumor mutational burden and neoantigen heterogeneity (clonal vs subclonal neoantigens) could mediate whether new mutations enhance or undermine effective anti-tumor T cell responses; these factors support the mechanism but are kept as secondary modifiers. 4) Endpoint choice: I selected therapy_response (response to pembrolizumab + cisplatin) because the path centers on treatment sequencing/combination effects and mechanistic modulation of immunotherapy efficacy by chemotherapy and TP53 status. 5) Deliberate exclusions: I excluded adding PD-L1/PDCD1 and other immune-checkpoint genes to the core hypothesis because they were not explicit in the provided path and would dilute mechanistic focus; I also avoided multiple genes in the primary claim to satisfy the Focusing Rule and to frame a testable mechanistic question rather than a correlative multivariate model?

Working Hypothesis

TLS-like lymphoid aggregates predict enhanced therapy_response to pembrolizumab plus cisplatin in muscle-invasive urothelial carcinoma in a TP53-dependent manner. In TP53 wild-type tumors, cisplatin-induced mutational burden increases clonal neoantigens that bolster TLS-driven anti-tumor immunity and improve therapy_response, whereas in TP53-deficient tumors, elevated subclonal neoantigen diversity diminishes TLS-driven responses, leading to lower therapy_response despite high mutational burden.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TP53
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53-deficient clones; pre-existing organized immune structures (TLS) should promote response to PD-1 blockade, yet clinical benefit of pembrolizumab+cisplatin is variable.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	cisplatin-induced mutational burden	Supporting modifier from knowledge graph

Secondary	tumor_mutational_burden (TMB)	Supporting modifier from knowledge graph
Secondary	neoantigen clonality (clonal vs subclonal)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective, multi-omic analysis of TLS status, TP53 mutation status, TMB, cisplatin-induced mutational signatures, and neoantigen clonality in cohorts treated with pembrolizumab+cisplatin, with therapy_response as the endpoint and interaction tests between TLS, TP53, and the mutational/neoantigen metrics;

Prediction 2: in vitro/in vivo models comparing TP53 wild-type and TP53-mutant urothelial cancer systems, with and without cisplatin exposure, to measure changes in TMB, clonal vs subclonal neoantigen burden, TLS formation markers, and T cell activity;

Prediction 3: mechanistic readouts including WGS/WES for mutational landscapes, neoantigen prediction pipelines, immunohistochemistry for TLS components (e.g., B cell follicles, CXCL13), single-cell T/NK cell profiling, and functional assays of TLS-associated T cell responses;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- cisplatin-induced mutational burden: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- tumor_mutational_burden (TMB): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.